

Keshav Aggarwal

PERSONAL DETAILS

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EDUCATION

JUL 2023 – MAY 2026	Ph.D. Astronomy – DAASE, Indian Institute of Technology (IIT) Indore (thesis submitted May 2026) Funding agency: Prime Minister’s Research Fellowship Collaborating agency: Space Physics Laboratory, Vikram Sarabhai Space Centre, Thiruvananthapuram Supervisor: Prof. Abhirup Datta <i>Exploring space weather and its effect on planetary atmospheres using radio occultation</i>
AUG 2021 – APR 2023	M.Sc. Astronomy – DAASE, Indian Institute of Technology (IIT) Indore Supervisor: Prof. Abhirup Datta <i>Study of the solar corona using the Indian Mars Orbiter Mission</i>

RESEARCH POSITIONS

OCT 2023 – PRESENT	Ph.D. Student Visitor , Space Physics Laboratory, Vikram Sarabhai Space Centre (ISRO), Thiruvananthapuram
OCT 2023 – MAY 2026	Prime Minister’s Research Fellow , DAASE, IIT Indore

WORKSHOPS, SCHOOLS, AND PRESENTATIONS

Mar 2026	“A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening” at 7th Indian Planetary Science Conference (IPSC 2026) , IIT Indore, India (Talk)
Feb 2026	“Estimating solar wind velocities using spacecraft radio signals” at National Space Science Symposium (NSSS 2026) , Umiam, Shillong, India (Talk)
Jan 2026	“Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki” at ISRO–ESA Workshop , Thiruvananthapuram, India (Poster)
Dec 2025	“Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki” at AGU Fall Meeting 2025 , New Orleans, United States (Poster)

- Dec 2025 *“Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki”* at **The Variable Sun, Thiruvananthapuram, India** (Poster)
- Sep 2025 *“Probing the Smiley-Sun using Akatsuki radio signals”* at **ASI Symposium 003, Astronomical Society of India, JECRC University, Jaipur, India** (Poster)
- Aug 2025 *“Studying the ‘Smiley Sun’ using the Akatsuki radio occultation experiment”* at **URSI Asia-Pacific Radio Science Conference (URSI AP-RASC 2025), International Convention Centre, Sydney, Australia** (Talk)
- Mar 2025 *“On the estimation of solar wind velocity using spectral characteristics of the probing radio signals: results from solar occultation measurements using Indian Mars Orbiter Mission”* at **6th Indian Planetary Science Conference (IPSC), IIT Roorkee, India** (Poster)
- Oct 2024 *“Results from radio occultation studies using Indian Mars Orbiter Mission”* at **6th URSI Regional Conference on Radio Science (URSI-RCRS), Graphic Era Hill University, Bhimtal, India** (Poster)
- Oct 2024 *“Results from radio occultation studies using Indian Mars Orbiter Mission”* at **3rd Indian Space Weather Conference, IIT Roorkee, India** (Talk)
- Oct 2023 *“Unveiling space weather and planetary atmosphere dynamics through Aditya-L1 and DISHA data integration”* at **2nd Indian Space Weather Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)
- Sep 2023 *“Retrieving sulphuric acid profiles of the Venus atmosphere from Akatsuki radio occultation data”* at **Venus Science Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)
- Aug 2023 **Strange New Worlds conference, Indian Institute of Science Education and Research (IISER), Pune**
- Feb 2023 **ISRO-ARIES Aditya-L1 Support Cell Workshop 3, Indian Institute of Technology (BHU), Varanasi**
- Jul 2022 **IIA Summer School 2022, Indian Institute of Astrophysics (IIA), Bengaluru**
- Jun 2022 **Introductory Summer School in Astronomy and Astrophysics 2022, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune**
- Mar 2022 **Spring 2022: ERS Post-Launch Data Challenge for JWST, Space Telescope Science Institute, Baltimore, and Heidelberg**
- Sep 2021 *“HR diagrams of star clusters”* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)
- Aug 2021 *“The Gaia telescope and the use of its data”* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)
- Feb 2021 *“Life in the Solar System, and exoplanets”* at **Nakshatra, Astronomy Club, IIT Indore, Indore, India** (Talk)

ACHIEVEMENTS

- Dec 2025 Recipient of the *AGU Student Travel Grant* for presenting a poster at AGU Fall Meeting 2025, New Orleans, USA
- Mar 2025 *Early Career Researcher Award for best poster* at the 6th Indian Planetary Science Conference, IIT Roorkee
- Feb 2024 Contributed to the Venus mission white paper '*Venus: the perfect laboratory for understanding interactions between space weather and planetary atmospheres*', submitted to ISRO
- Oct 2023 *Prime Minister's Research Fellowship*, Ministry of Education, Government of India

INTERNSHIPS

AUG 2022 – SEP 2023	Researcher in AI and Orbital Mechanics HR Coordinator at ERETS Space London, London (remote) Developed AI/ML and orbital mechanics algorithms for satellites. Handled hiring for the team.
FEB 2022 – AUG 2022	Data Analyst Project Coordinator at Armstrong Space Australia, Melbourne (remote) Built pipelines for acquiring and processing space debris data. Space debris removal by laser ablation. Catalogues of debris between 1 and 10 cm.

FIRST-AUTHOR PUBLICATIONS

1. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Manikantan, R., Sharma, A., Imamura, T., *A study on the contribution of the interplanetary medium in radio occultation experiments*, **Icarus**, 2026, [10.1016/j.icarus.2026.117291](https://doi.org/10.1016/j.icarus.2026.117291).
2. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Sharma, A., *Plasma turbulence in the lunar environment across solar wind and magnetotail conditions: Observations from Chandrayaan-2 Radio Science experiment*, **Journal of Geophysical Research: Planets**, 2026, [10.1029/2026JE009957](https://doi.org/10.1029/2026JE009957).
3. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Imamura, T., Ando, H., *A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening*, **Monthly Notices of the Royal Astronomical Society**, 2026, [10.1093/mnras/stag554](https://doi.org/10.1093/mnras/stag554).
4. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Roopa, M. V., Imamura, T., *A generalized method for estimating solar wind speeds and densities using spectral broadening for a Kolmogorov turbulence spectrum*, **Advances in Space Research**, 2026, [10.1016/j.asr.2026.03.059](https://doi.org/10.1016/j.asr.2026.03.059).

5. **Aggarwal, K., Choudhary, R. K., Datta, A., Imamura, T.,** *On the estimation of solar wind velocity under varying solar activity conditions using Akatsuki measurements*, **Monthly Notices of the Royal Astronomical Society**, 2025, [10.1093/mnras/staf1305](https://doi.org/10.1093/mnras/staf1305).
6. **Aggarwal, K., Choudhary, R. K., Datta, A., Roopa, M. V., Dai, B. K.,** *Insights into solar wind flow speeds from the coronal radio occultation experiment: Findings from the Indian Mars Orbiter Mission*, **The Astrophysical Journal**, 2025, [10.3847/1538-4357/adb627](https://doi.org/10.3847/1538-4357/adb627).
7. **Aggarwal, K., Datta, A., Choudhary, R. K.,** *Studying the 'Smiley Sun' using the Akatsuki radio occultation experiment*, **Proceedings of the 7th URSI Asia-Pacific Radio Science Conference (AP-RASC 2025)**, 2025, [10.46620/ursiaprasc25/ahct9245](https://doi.org/10.46620/ursiaprasc25/ahct9245).
8. **Aggarwal, K., Choudhary, R. K., Datta, A.,** *Results from radio occultation studies using Indian Mars Orbiter Mission*, **Proceedings of the 6th URSI Regional Conference on Radio Science (RCRS 2024)**, 2024, [10.46620/ursi_rsrc24/1017qon9346](https://doi.org/10.46620/ursi_rsrc24/1017qon9346).